**UNIVERSIDAD DE CHILE**
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**ESCUELA DE SALUD PÚBLICA**

# SUBSTANCE USE TREATMENT AND POST-TREATMENT OUTCOMES, 2010-2022: A nationwide retrospective data linkage cohort study

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**PERFIL DE TESIS PARA OPTAR AL GRADO DE DOCTOR EN SALUD PÚBLICA**

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**Abstract**

There is an increasing prevalence of substance use disorders (SUD) in Chile added to a lack of systematic practices for evaluating treatments. Observational data can be used to assess the effect of SUD treatment on post-treatment outcomes.

This study aims to investigate the effect of the length of SUD treatment and post-treatment outcomes: readmission to SUD treatment (i), contact with the justice system (ii), and mortality (iii) among Chilean SUD adult patients from 2010-2022. The study further explores the potential heterogeneity of this effect by treatment setting (residential, intensive-ambulatory, and ambulatory) and treatment outcome (completion, non-completion). The study will use a nationwide retrospective data linkage cohort design of patients in adult treatments recorded by the National Service of Drug and Alcohol Use from 2010-2022, linked with data from the Prosecutor's Office, and the Ministry of Health.

The findings of this study are expected to contribute significantly to the understanding of SUD treatment effectiveness and inform policy and practice in Chile's public health sector.

**SUBSTANCE USE TREATMENT AND POST-TREATMENT OUTCOMES, 2010-2022: A nationwide retrospective data linkage cohort study**

***SUD seen by PH, World estimates***

Decades of research on alcohol and substance use disorders (SUD) have shown that these phenomena are complex, encompassing multiple dimensions beyond purely biochemical or biomedical neurological domains (1–3). Evidence shows that SUDs result from social and psychological conditions interacting with biological factors and complex biographical processes (1,2), in which pragmatic, population and environmentally focused interventions tend to produce better outcomes (3,4).

As such, SUDs are a major public health issue. They are linked to various health problems and disabling conditions, such as neuropsychiatric and physical disorders (5,6) and social risk factors (e.g., risky sexual behavior, involvement in traffic accidents, or engaging in violent incidents) (7–9). Also, they are one of the leading causes of mortality and morbidity worldwide, accounting for 0.52% (95% CI= 0.4%, 0.55%) of global mortality and 1.38% (95% CI= 1.17%, 1.60%) of disability-adjusted life years (DALYs) (10–12). Furthermore, SUDs significantly contribute to early deaths and disabilities across the Americas. These disorders (alcohol aside) were the ninth leading causes of DALYs in the region in 2019, with a rate of 782.2 DALYs per 100,000 individuals (13).

***Health burden- Chile***

Chile is one of the countries with the largest morbidity burden of psychiatric diseases globally, with alcohol dependence as one of the leading causes of disability in life-adjusted years (14). Also, it is one of the countries with the largest average alcohol consumption in the Americas (15). These antecedents may partly explain why alcohol and SUDs are the third and fourth lifestyle and behavioral sanitary objectives of the Health National Strategy 2021-2030 (16).

In the last decade, substance use has increased significantly and remained relatively stable in Chile, although high since then. So much so that until 2016, Chile showed the highest prevalence of cocaine, crack, and marijuana use among school students in the Americas (17). Assuming that SUDs are relatively constant among people who showed SUDs, it is likely that the number of people that will require treatment as adults will increase in the near future. Already contributing to the current statistics and according to the last publicly available National Drug Survey in the General Populations conducted in 2020, the National Service of Drug and Alcohol

Use (SENDA) estimated that 2.9% of the population (12-64 years) met the criteria for abuse or dependence on alcohol, marijuana, cocaine, or cocaine-paste (18).

***Treatment***

Implementing treatment programs has been one way to tackle these increasingly demanding needs (7,19). Treatment for SUD aims to reduce problematic substance use, lessen the severity of withdrawal symptoms and related behaviors, improve health and functional status, and modify contextual risk factors for these behaviors (7,20).

Despite a considerable agreement on the potential benefits of treatment, consensus regarding how to measure them remains elusive and often may lead to inconsistent results (21). Among the several ways to measure treatment, treatment retention and completion are the more studied ones, emphasizing its different but complementary properties/aspects (21).

*Length of stay*. The length of stay in treatment can be used here as a synonym for adherence or retention. More importantly, treatment characteristics, such as frequency and intensity, may not be as important as the time spent in such a program (22). Nevertheless, there is an open debate about the duration of treatment related to positive post-treatment outcomes. On the one hand, evidence has highlighted 90 days as a minimum duration for obtaining the benefits of treatment in terms of facilitating meaningful behavioral changes and other outcomes (23). On the other hand, six months has also been associated with better outcomes in some treatment modalities (24), and even some studies have found that 30 days can have positive effects (25). In addition, several studies have discussed a continuous but not necessarily linear link between post-treatment outcomes and treatment duration. They propose an optimal 'Goldilocks' duration zone, which strikes a balance to maximize efficacy and minimize patient dropout (26), allowing us to define when treatment is “enough” (27) rather than a “dose-effect” (in which patients get diminishing returns when reached a certain amount of treatment sessions) (28), or a linear dose-response effect in which “more is better” (29).

*Treatment completion*. Another useful measurement is treatment completion, which means achieving goals defined by the treatment center professional team and those previously agreed upon with the patient (vs. leaving treatment “against medical advice” or for “disciplinary” reasons) (30). Noncompletion is related to social and health-related outcomes (31). Readers should know that far from impulsive behavior, dropout may be due to several motives, including social support and mental health burden (32).

Zarkin and colleagues position a non-reductionist definition of treatment:

...“*although many studies suggest that treatment success depends largely on length of stay, it may also be that success is closely related to other aspects of treatment, such as whether a patient completes a planned treatment regimen as prescribed by the program*”(33) (p. 262)

Hence, we may consider both measurements to study treatment adequately.

Another source of heterogeneity worth considering is when defining treatment modality. These vary from 24-hour all-week residential care to less intensive ambulatory programs. Treatment assignment may be endogenous to patients’ clinical and psychosocial characteristics in most cases, with residential treatments directed to more severe patients in terms of substance use patterns, health-related and sociodemographic risk factors (7,34). In Chile, residential care is intended to last a year and offers 5-7 hours daily with at least five interventions per week. Ambulatory treatments, lasting 6-8 months, demand 2 (for basic) to 5 (for intensive settings) days per week attendance with sessions lasting 2-6 hours and up to 4 interventions (35,36). These differences should be accounted for when studying the link between treatment and post-treatment outcomes.

There is a dimension that is not included, and its measurement in Chile is very recent (37). Although patient-based experiences may be one way to measure treatment, these are often not included in many studies, and literature has shown they are closely related to the abovementioned measurements (38).

***Effectiveness, are substance use treatments beneficial? On what?***

Having defined how treatment can be measured, it is necessary to elucidate how we will assess the effect of these treatments and how they are measured in post-treatment outcomes. Evidence has found that treating SUDs is associated with reductions in substance use, psychological symptoms, and psychosocial functioning in general, including criminal behavior and employment (39–41).

A traditional approximation would focus on the measurement of recovery or long-term abstinence. Still, untreated remission from SUDs is predominant, even without a history of treatment, while others often report more than one recovery attempt (42,43). For instance, there is broad support for addressing SUD as a chronic condition, thus acquiring a chronic health perspective when assessing effect indicators (3,44). Furthermore, there is consensus on evaluating multiple longitudinal outcomes, as there is no “gold standard” outcome (45). Hence, it is reasonable to expect to work with data with a panel or time-to-event structure.

Studies hitherto examined the relationship between SUD treatment and various post-treatment outcomes beyond relapse. These include psychosocial outcomes, such as a

greater likelihood of employment and decreased contact with the criminal justice system (31,46–48). They also cover SUD reduction patterns and diminished probabilities of readmission (also referred to as re-presentation) to SUD treatment (49–52), and even reduced mortality rates (47,53,54). However, most existing research explore these outcomes in isolation.

***Research gaps: Scarce research in the global south, Difficulties studying treatments, Effectiveness evaluations are scarce***

However, and to our knowledge, most of the research on the effect of treatment in several post-treatment outcomes has been conducted in the global north, with different epidemiological, social, and substance use contexts and health systems (37,40,55–59). Thus, extrapolating these results to the Latin American and Chilean context can be misleading, given their focus is on less prevalent substances in the region, such as heroin or other opioid-related substances. Instead, cocaine paste base is the most widespread form of smoked cocaine and the second most prevalent substance behind alcohol (60–63) and younger people in treatments (alcohol aside) (64). In particular, research conducted by the Chilean Budget Office underscores that SENDA currently lacks robust and systematic practices for evaluating treatments (59).

***A window of opportunity [tentativo]***

The lack of centralized data systems in several Latin American countries is a critical limitation for research and program management. However, Chile has overcome this difficulty due to implementing a centralized data registry system, SISTRAT (named for its Spanish acronym), which became operational in 2010 (65), and is one of Latin America's oldest and most advanced SUD treatment monitoring systems (56). This system registers all patients admitted to public adult (18+ years old) SUD treatment programs in centers affiliated with SENDA's network. The Chilean public health system covers 81% of contributors, possibly reaching 85% by 2023 (66). The coverage of SUD treatments has increased, registering approximately 16,500 adults yearly (67). It contains relevant patient information (i.e., sociodemographics, health status, and substance use antecedents). In this context, SISTRAT data offers an opportunity to generate local evidence on the association of treatment and post-treatment outcomes of SUD patients.

Additionally, Chile has monitored patients' post-treatment outcomes since the implementation of the Treatment Outcomes Profile in 2015. This validated questionnaire is administered to patients upon admission, every three months during treatment, and subsequently for post-treatment follow-ups up to six months after treatment. It extensively

evaluates various outcomes, including patterns of drug use, crime-related behaviors, and occupational status (68).

Also, Chile has an exceptional mortality system, which complies with many international standards. These events are registered in the Department of Statistics and Health Information, including about 100,000 cases annually, with more than 99% of all cases with a medical death certificate (69–71).

Another valuable information system is the Prosecutor's Office (PO) data, managed by the Prosecutor's office, which oversees the investigation, prosecution, and adjudication of crimes (72). Chile boasts significant governmental transparency, encompassing government data and the right to information, placing Chile in 18th of 102 countries according to the World Justice Project Open Government Index (73,74).

We believe that all of these sources have the potential to offer insights into the effect of SUD treatment and post-treatment outcomes, as linked administrative databases may have some advantages over other longitudinal data collection methods, such as no loss to follow-up and continuous data collection (75). However, their combined use in population-level research is unknown to the writer of this proposal.

***Observational studies to assess treatment effects***

Let us now consider how to study the effect of SUD treatment. Despite a paucity of randomized designs on the effect of treatment on post-treatment outcomes, usually regarded as the pinnacle of evidence-based studies (but often not affordable to monetary and logistic constraints) (51), national data routinely collected in Chile can be used to evaluate local- and national-level interventions, such as treatments for SUDs, in terms of their effect on post-treatment outcomes. In this regard, observational data have the potential to answer causal questions by estimating the effect under a hypothetical randomized control trial in the absence of true experimental designs (45,76,77). Observational studies provide practical insights into the effectiveness of interventions in real-world conditions, unlike clinical trials that focus on whether they work under ideal conditions (known as 'efficacy') (23,78). Therefore, when conducting a clinical trial isn't feasible, these studies can offer a valuable opportunity to measure how well treatments work in normal circumstances (79).

One of the main requirements is the stable-unit-treatment-value assumption, which leads us to think about the assignment mechanisms behind data to define and identify causal associations, encompassing consistency (i.e., whether everyone received a similar level of

treatment, no interference or hidden variations in treatment) and positivity (i.e., both exposed and non-exposed individuals across all confounding levels) (80). Consequently, design-based approaches should overcome other methodological challenges to produce valid estimates of the effect of treatment. One of the main limitations is the uncontrolled confounding of the associations between treatment and outcomes (conditional exchangeability) (81). Another should be the competing risk of other outcomes, for example, of deceased patients, which could lead to survival bias (82). In the context of research on SUD treatment and applying a survival framework, censoring (left, right, and interval), time-scale (e.g., time zero, calendar time), frailty (over longitudinal, repeated and lag-dependent measurements), and time-varying confounders (or time-dependent factors) and associations (time-varying coefficients with non-proportional hazards) have to be considered to study the effect of SUD treatments on post-treatment outcomes (transient or absorbing) (83,84).

Given the lack of knowledge in the country, What is the distribution of the time to different outcomes of patients in Chilean SUD adult treatments between 2010-2022?, What is the functional relationship between the length of stay in SUD treatment and post-treatment outcome of patients in Chilean SUD adult treatments between 2010-2022? Whether an increase in SUD treatment length will result in better post-treatment outcomes? Is this relationship between SUD treatment length and post-treatment outcomes heterogeneous on the different treatment modalities and outcomes?

**Objectives**

To analyze the effect of the length of SUD treatment on post-treatment outcomes such as readmission to SUD treatment (i), contact with the justice system (ii), mortality (iii), and employment (iv), and then examine how this effect might vary by treatment setting (residential, intensive-ambulatory and ambulatory) and treatment outcome (completion, non-completion), among patients in Chilean SUD adult treatments from 2010-2022.

***Specific objectives***

1) To describe the sociodemographic (e.g., sex), substance use patterns (e.g., substance use frequency), health (e.g., psychiatric comorbidity), treatment (e.g., public/private status of treatment center), and related to the geographic location (e.g., poverty index of the commune or municipality of residence) characteristics of patients in Chilean SUD adult treatments between 2010-2022.

2) To estimate the time-to-event of readmission to SUD treatment (i), contact with the justice system (ii), mortality (iii), and employment (iv) of Chilean SUD adult treatments between 2010-2022 individually.
3) To explore how this time-to-event varies by treatment setting (residential, intensive-ambulatory and ambulatory) and treatment outcome (completion, non-completion) individually, among patients in Chilean SUD adult treatments from 2010-2022.
4) To estimate the effect of SUD treatment on post-treatment outcomes jointly (i.e., accounting for competing risks), fixed and time-varying confounders and time-varying associations, among patients in Chilean SUD adult treatments from 2010-2022.

**Methodological Approximation**

***Design & Participants***

Observational study using data from a nationwide retrospective linked cohort of administrative data of patients with an ongoing SUD treatment between 2010-2020 in Chile. Preliminary analyses made by the writer of this proposal estimate that 106,534 patients underwent treatment in this period. Data were collected using information from electronic medical records, which contain data from self-report questionnaires at treatment entry and information from follow-up interviews with the clinical team.

***Sample size***

We estimated the proportion of patients we expect to experience the rarest event: mortality (85). Mortality rates were 7.43 per 1,000 by 2021 in the general population (86), although this proportion might seem conservative given that mortality rates tend to be greater in people with SUDs (87). We performed a sample size calculation using the "powerSurvEpi" package in R, using the approximation of Hsieh and Lavori (88): Presuming a prevalence of 0.743%, a 2-tailed type I error $\alpha$ of 0.005 instead of 0.05 to allow correction for multiple comparisons over the levels of a covariate (i.e., treatment setting), a hazard ratio of order 0.99 (i.e., a hypothetical scenario in which we would be able to detect a protective effect of 1/0.99= 1.01% of the length of treatment stay, given that it is a continuous variable, thus effects might be short), a standard deviation of 176.5 according to a previous study in the Chilean population (89), and assuming that days in treatment might not be independent of other covariates ($R^2$= 0.80), 3,181 observations will be needed to obtain a power of 90%. It is important to note that 16,500 enter treatment annually, and deaths are less frequent than other post-treatment outcomes. Hence,

sample size requirements are expected to be lower for the rest of the outcomes. We expect to account for competing risks in sample size estimates in the future.

### *Variables*

A part of the research will be advocated into defining variables and their role.

***Exposure***: Length of treatment (time from baseline treatment admission to treatment outcome); ***Outcomes***: Time to contact the justice system after treatment (committing an offense that led to a condemnatory sentence), readmission to SUD treatment, self-reported occupational status (improved) after treatment (interval-censored at 3 and 6 months after treatment), and time to death. The time will be measured starting from the end of treatment. ***Covariates***: Sociodemographic (e.g., sex), substance use patterns (e.g., substance use frequency), health (e.g., psychiatric comorbidity), treatment (e.g., public/private status of treatment center), related to the geographic location (e.g., poverty index of the commune or municipality of residence), and previous events (e.g., adolescent treatment, contacts w/ justice system before treatment admission). Effect modifiers: Treatment modality (ambulatory/ intensive ambulatory/ residential) and Treatment outcome (completion or non-completion).

### *Analytic Plan*

We will link data via a deterministic linking method. This process involves encrypting the patient's national identification number (RUN), which will be applied across all the datasets included in this study. We will carry out the encryption process with an offline web application created for the ANID-FONDECYT project 1191282 led by Dr. Castillo-Carniglia, capable of standardizing RUNs and converting them into a unique hash code using the MD5 (Message-Digest) algorithm, a widely used tool in security protocols to encrypt information into a 128-bits hash-code and improved by coding it with a salt-key (user-chosen additional characters for further encryption) (90).

The primary cohort was sourced from the SENDA adult substance use treatment surveillance system SISTRAT. Mortality datasets contain basic demographic information, the date and municipality of the death, and the underlying and external cause of death according to ICD-10. PO database includes data on official criminal charge records and sociodemographic information (i.e., sex, birth date). Finally, The TOP questionnaire database is a resource that contains sociodemographic information, substance use patterns, and basic employability information of patients followed up after treatment, depending on the number of people followed after treatment from the date of implementation (i.e., 2015).

Another critical part of the process involves data exploration and cleaning. Considering all the variables available in the databases, we will explore the data focusing on missing data, variance, and other descriptive measures.

After this process, we will carry out a data normalization process. Whenever possible, we will standardize variables, adequately label fields, and correct data integrity issues (e.g., typographical errors in dates, automation bias, variations in name spelling or form, and aberrant values with |1,5 |*IQR following Tukey's criteria). We will conduct consistency analyses between the databases before merging (i.e., demographic data should be the same in all sources for an individual wherever available).

Once certain key variables are cleaned, we will merge datasets from SISTRAT, DEIS, and PO to create a single dataset for statistical analysis. The datasets will be transformed, recoded, deduplicated, standardized, and appended into a single database in a wide format (i.e., one patient by row) (91).

Discrepancies will be resolved by collapsing entries with non-significant referrals followed by a similar treatment in a short period, through logical imputation within users on time-invariant variables using heuristics, decision trees, and schemes upon IF-THEN statements. If the abovementioned techniques do not distinguish treatments, we will use more advanced tools such as imputations, probabilistic matching, or deduplication (92).

As part of the objectives of the study, we will construct a Directed Acyclic Graph (DAG) to identify a minimum set of confounders, colliders, and mediators according to the literature and the back-door criterion will be applied to identify a minimally sufficient set for confounding adjustment (a preliminary DAG is offered in the Annex) (93). We will conduct descriptive analyses, such as frequency distribution and rates of our exposure variables, stratifying these by treatment setting and treatment outcome. Incidence rates per 1,000 person-years and Kaplan-Meier curves stratified by treatment modality on the time-to-each-outcome will be explored, also analyzing each univariate survival distribution.

Next, we will conduct several tests (e.g., Schoenfeld, log-ranked-based, supreme, etc.) of violations of the proportional hazards assumption (84). Missing data will be imputed to minimize potential bias, whether possible to do so. Covariate-adjusted transition intensities will be estimated through individual-level continuous time state transition models to complement time-to-first-event methods that do not account for time-varying coefficients and competing risks (94–97), also robust to misspecification and other methodological challenges already

asserted (e.g., limits of detection of SUD treatment, lack of common support, treatment confounder feedback) (98,99). Analyses will be performed in R & Stata softwares (100,101).

***Ethical concerns***

One of the most controversial research fields is using data based on individual health-related information recorded against explicit consent for the study objectives, considering that we will work with individual data. Population-based research requires population-based analyses as long as it is used for the common good and intervention in the context of uncertainty and does not attempt privacy and confidentiality through identifiability (102). Consent is only one factor and can even be considered one of the least critical ethical factors of clinical research (103). The data management plan will include the data collection process, and data curation will be registered and described in an accessible and transparent manner. Privacy protections in the event of (un)intended re-identification of individuals (104).

Potential patient confidentiality and anonymity breaches pose serious risks, so we have devised a data management and storage protocol. No personally identifiable information will be included in the datasets, with only encrypted unique identifiers available. Data will be stored securely in HIPAA-compliant cloud storage (i.e., Google Drive) with restricted access. Finally, results will be presented in aggregated form to protect individual identities (105).

**Work plan and schedule**

We can expect data acquisition, management (data cleaning, standardization, and linkage), and analysis to be time-consuming, based on the prior experience as a full-time analyst in FONDECYT project No. 1191282 led by Dr. Castillo-Carniglia using secondary information. We will delve into the transformation and processing of data following a longitudinal perspective of nested events by subjects. Additionally, we will devote significant time to discussing and implementing knowledge of causal inference involved in the analysis.

Given the writer of this proposal is currently working part-time as a young researcher in the ANID—Millennium Science Initiative Program, no. NCS2021_003, led by Dr. Castillo, we have managed to access several data sources.

- Regarding patients in SENDA treatment programs, there is a collaboration agreement within the framework of the project FONDECYT Regular N° 1191282 and NCS2021_003. The writer of this proposal is already working on the SENDA databases.
- Regarding PO data, we already accessed records of patients in SUD treatments from 2010-2019. We will request an update to get data up to 2022.

- Regarding mortality data, a signed Institutional Collaboration Agreement of Universidad Mayor, with Dr. Alvaro Castillo as the principal investigator of this proposal, and the Ministry of Health MINSAL has already passed the final administrative act, so we are waiting to begin the data acquisition process (see Annex section for more details).

***Thesis format*:** Two research papers will be sent to journals indexed in the Web of Science. We aim to present this study at one international conference, such as the Society for Epidemiologic Research or similar, and possibly in scientific community activities organized by Universidad de Chile or other national institutions.

Table 2. Expected timeline

| | **2023 (quarters)** | | | | **2024 (quarters)** | | | | **2025 (quarters)** | | | | **2026** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Q11 | Q12 | Q13 | Q14 |
| Acquiring data | | | | | | | | | | | | | | |
| Data cleaning, linkage and standardization | | | | | | | | | | | | | | |
| Analysis, writing and paper #1 preparation | | | | | | | | | | | | | | |
| Analysis, writing and paper #2 preparation | | | | | | | | | | | | | | |
| Presentation in scientific meetings | | | | | | | | | | | | | | |
| Compile findings and research backgrounds | | | | | | | | | | | | | | |
| Thesis defense | | | | | | | | | | | | | | |

**Originality and Relevance**

Much of the work FONDECYT Regular No. 1191282 led by Dr. Castillo-Carniglia attempted to assess the relationship between SUD treatments and multiple outcomes (i.e., hospitalizations, deaths). However, much of the project was truncated due to the COVID-19 pandemic, which compromised DEIS's capabilities to comply with data requirements. Given the actual advances in the agreement, we are confident that we will acquire the necessary data in this instance.

Evidence indicates mistrust in SUD treatments in Chile, along with beliefs that most SUDs can resolve themselves and may not benefit from treatment (106). Given the lack of evidence on treatment effects on post-treatment outcomes (65), there's a pressing need for studies to determine the 'added value' of SUD treatments in our country. This need is likely heightened by the treatment access gap for SUD: 1 out of 10 problematic substance users received treatment for substance use disorders (107). Even though these estimates are not entirely comparable, they are greater than recent global estimates, suggesting that 1 in 5 people receive treatment (excluding alcohol) (64).

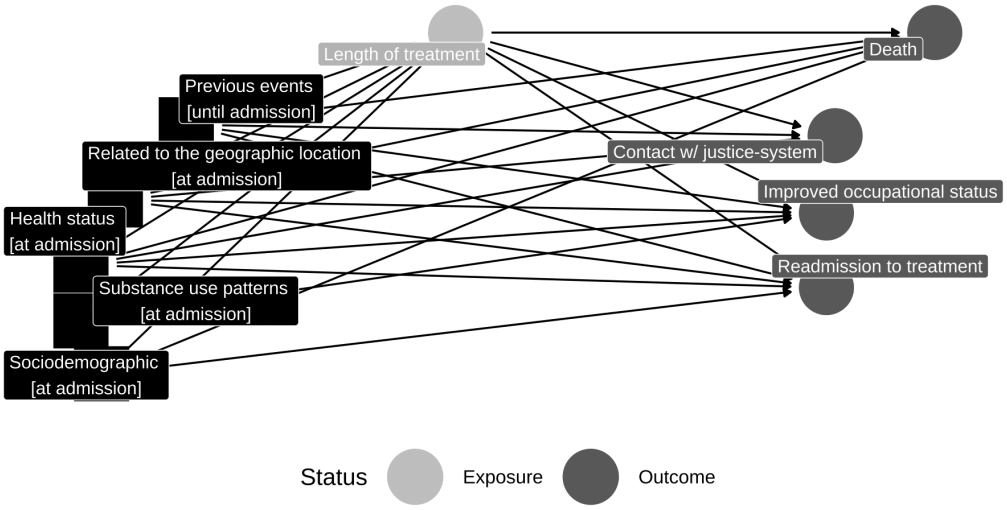
Also, this thesis is based on the linkage of databases, which has been a challenge for public health data science and has been highlighted recently by our country's different

institutional and academic key actors in public health and data science (108). This thesis may contribute to generating data linkage practices and products that enable the creation of future research, publications, and applications in this field.

A key strength of this proposal lies in its broad geographical scope, spanning multiple jurisdictions, including various regions and cities. This broad reach bolsters confidence that local factors, such as policies at individual centers, will not influence the results. Furthermore, we could effectively examine the effect of SUD treatment on intricate time-dependent outcomes with time-varying associations by employing an extensive follow-up period of over a decade. This line will contribute to the existing literature on the effects of SUD treatments (29,109). The results will impact public health, as they will inform about the benefits and potentials of the treatment system (one of the objectives of the National Drug Strategy 2021-2030) (107) through reproducible code and freely accessible, aggregated, and interactive data. The study will be open and publicly available for review and scrutinization by peers and anyone interested in the project through GitHub (link1 and in the project's GitHub: link2), mainly for transparency, future update, and replication to different treatments (e.g., rules for logical imputation or deduplication). The study will advance the application of big data in health research and decision-making, expecting to contribute to opening new frontiers in public health research on SUDs in Chile (110). Consequently, the study's objectives align with public health strategies in Chile and sanitary objectives, such as managing interoperable systems with data from different sources, thus generating information to benefit public health, particularly in marginalized populations (i.e., with structural social disadvantages, neighborhood conditions) that bear the brunt of a lack of sufficiently evidence-based programs (16). This information has been used for internal, administrative, and budgetary purposes (e.g., renewing treatment center agreements with SENDA) but has rarely been used as an epidemiological monitoring and surveillance mechanism. The results of this thesis can be used as an evaluation tool to inform current and subsequent public health planning on SUD treatments (111) and the allocation of increasingly scarce resources that compete with policies with punitive rhetoric (112). Additionally, this thesis may contribute to the knowledge gap in SUD treatment from the Global South (different epidemiological contexts and health systems) regarding the best way to measure treatment and its results, constituting an opportunity to generate local evidence from which other countries with similar characteristics can learn from the Chilean case (37).

## Annex

Annex Figure 1. Dynamic Acyclic Graph




Source. Prepared by the authors

Annex Figure 2. Screen capture of the agreement

FRR/PBS/RZM/YVB/CHT

**APRUEBA CONVENIO DE COLABORACIÓN ENTRE LA SUBSECRETARIA DE SALUD PUBLICA DEL MINISTERIO DE SALUD Y LA UNIVERSIDAD MAYOR.**

**EXENTO N°**

**SANTIAGO,**

**VISTO:** Lo dispuesto en el D.F.L. N°1, de 2005, del Ministerio de Salud, que fija el texto refundido, coordinado y sistematizado del Decreto Ley N°2.763, de 1979, y de las leyes N° 18.933 y N°18.469; en el Decreto Supremo N°136, de 2004, Reglamento Orgánico del Ministerio de Salud, Decreto Supremo N°28 de 2009, del Ministerio de Salud y en la Resolución Nº 7 de 2019 de la Contraloría General de la República;

**CONSIDERANDO:**

**1º** Que, al Ministerio de Salud le compete la función que corresponde al Estado de garantizar el libre e igualitario acceso a las acciones de promoción, protección y recuperación de la salud y de rehabilitación de la persona enferma; así como coordinar, controlar y, cuando corresponda, ejecutar tales acciones.

**2º** Que, la Universidad Mayor declara que es una institución de educación superior que tiene como preocupación permanente la de impulsar y fortalecer la investigación y el desarrollo científico, mediante un enfoque interdisciplinario, priorizado y global, para atender las necesidades y demandas de los sectores sociales, culturales y productivos del país y que, en el ámbito de transferencia tecnológica e innovación, dispone de unidades de investigación, docencia y extensión en estas temáticas.

**3º** Que, en el marco del concurso Núcleos Milenio en Ciencias Sociales, año 2021 de la agencia Nacional de Investigación y Desarrollo, la Universidad Mayor patrocinó la postulación del Dr. Álvaro Ernesto Castillo Carniglia, quien se adjudicó, en calidad de Director, el proyecto de interés público denominado "Millenium Nucleus for the Evaluation and Analysis of Drug Policies [Núcleo Milenio para la evaluación y análisis de políticas de drogas (nDP)]" (en adelante el "Proyecto"), Código #NCS2021_003. El proyecto tiene como finalidad generar conocimiento basado en evidencia y recomendaciones sobre políticas de drogas en Chile

**4º** Que, con fecha 21 de abril de 2023, la Subsecretaria de Salud Pública y la Universidad Mayor, suscribieron un Convenio de Colaboración, cuyo objetivo es la cooperación y el intercambio de información entre ambas instituciones, que aporte al desarrollo de políticas públicas multisectoriales y genere un avance para proveer mejores mecanismos para la toma de decisiones, diseño y desarrollo de políticas públicas.

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